

HYDROGENATED NBR (HNBR) PATENT LANDSCAPE: SYNTHESIS, LOW-MOLECULAR-WEIGHT CONTROL, AND PERFORMANCE-ORIENTED FORMULATIONS

1. ABSTRACT

This report synthesizes a set of primary patents relevant to hydrogenated nitrile butadiene rubber (HNBR), with emphasis on how precursor NBR is hydrogenated, selectively degraded, or formulated to deliver improved processability and end-use performance. The target material is HNBR defined by partial to near-complete saturation of the butadiene-derived backbone while retaining nitrile functionality, with key attributes including residual unsaturation, molecular weight, Mooney viscosity, gel content, and retention of mechanical properties after hydrogenation. Across the selected patents, two major synthesis directions emerge. First, several documents focus on direct hydrogenation or simultaneous metathesis/hydrogenation of NBR using homogeneous ruthenium/osmium catalyst systems, often in the presence of hydrogen and sometimes a co-olefin, to reduce molecular weight and narrow molecular-weight distribution in a single step. These routes aim to preserve polymer microstructure while lowering Mooney viscosity and residual double-bond content. Second, other patents address downstream formulation and processing of HNBR, including peroxide-cured compositions, microgel-containing vulcanizates, belt compounds, and residue-removal processes that improve manufacturability and purity after metathesis/hydrogenation. The landscape shows repeated use of solution-phase processing, catalyst tuning, and additive control to balance hydrogenation efficiency against gelation, corrosion, and catalyst residue issues. Representative formulation trends include use of unsaturated carboxylic acids/salts, peroxides, microgels, carbon black or silica fillers, and polyfunctional co-crosslinkers to improve hardness, low-temperature flexibility, LLC resistance, and high-temperature durability. Overall, the patents collectively indicate that successful HNBR synthesis depends on controlling backbone saturation and molecular weight while minimizing side reactions and preserving processability for seals, belts, hoses, and related elastomeric parts.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis itself; it discloses peroxide-vulcanizable compositions based on hydrogenated nitrile rubbers combined with unsaturated carboxylic acids/salts and specific microgels. The process emphasis is on compounding and vulcanization of HNBR to improve high-temperature mechanical properties and demoulding behavior. - Hydrogenated nitrile rubbers: degree of hydrogenation 95 to 100% — Claim 1 - Hydrogenated nitrile rubbers: Mooney viscosity (ML 1+4 @ 100°C) 10 to 120 MU — Claim 1 - Unsaturated carboxylic acids/salts: 1 to 100 parts by weight per 100 parts rubber — Claim 1 - Peroxide: 0.2 to 8 parts by weight per 100 parts rubber — Claim 1 - Microgel glass transition temperature: less than or equal to -20°C — Claim 1 - Microgel content: up to 60 parts by weight per 100 parts rubber — Claim 2 - Microgel glass transition temperature: less than or equal to -50°C — Claim 3 - Hydrogenated nitrile rubbers: Mooney viscosity 15 to 100 MU — Claim 4 - Hydrogenated nitrile rubbers: degree of hydrogenation 50 to 100% — Detailed description - Hydrogenated nitrile rubbers: degree of hydrogenation 85 to 100% — Detailed description - Operating temperature: 130°C — Summary of invention - Hot-air storage temperature: 150°C — Summary of invention

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance Although not a synthesis patent, it is highly relevant because it defines HNBR performance requirements and shows how peroxide-cured HNBR formulations can be tuned with microgels and unsaturated carboxylate additives. It provides practical formulation constraints for high-temperature HNBR applications such as belts, hoses, and gaskets.

Patent 2

- Patent Number: US9243683B2
- Patent Title: Toothed belt
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is a belt-article patent rather than an HNBR synthesis patent. It discloses a toothed belt whose rubber portion contains hydrogenated nitrile rubber and a zinc methacrylate-dispersed polymer alloy, with plasticizer balance tuned for low-temperature flexibility and stiffness. - Tooth pitch: 3 mm — Example 1 - Belt length: 339 mm — Example 1 - Belt width: 15 mm — Example 1 - Dioctyl sebacate to trimellitic acid ester ratio: 1:19 to 1:1 — Claim 1 - Dioctyl sebacate content: less than 4 mass% of rubber portion — Claim 1 - Total dioctyl sebacate + trimellitic acid ester: 4 to 8 mass% of rubber portion — Claim 1 - Bound acrylonitrile content: not more than 19 mass% — Claim 2 - Iodine value: not more than 15 mg/100 mg — Claim 2 - Mooney viscosity at 100°C: not less than 60 — Claim 2 - HNBR content in rubber portion: 15 to 50 mass% — Claim 2 - Welding angle: 10 degrees — Example 1

Relevant Experimental Evidence - Example 1: A toothed belt was fabricated using the specified HNBR-containing rubber portion, nylon-66 canvas, and carbon core wire; belt dimensions were given. - Example 2-5: Similar belts were prepared using alternative rubber portion compositions to compare performance. - Comparative examples: Variations in plasticizer ratio and rubber composition were used to show the importance of the specified DOS/TMLA balance for cold resistance and stiffness.

Technical Relevance This patent is relevant because it shows how HNBR is incorporated into high-performance belt structures and how HNBR composition parameters such as bound AN, iodine value, and Mooney viscosity affect final belt properties. It is useful for understanding downstream requirements that HNBR synthesis must satisfy.

Patent 3

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a catalyst system for metathesis reactions, including nitrile rubber metathesis, using a ruthenium- or osmium-based catalyst together with an alkaline earth metal chloride. The key concept is catalyst activation/acceleration for polymer metathesis, which can be applied to NBR degradation before hydrogenation to HNBR. - Salt:catalyst weight ratio: 0.01:1 to 10000:1 — Claim 1 - Catalyst metal: transition group 6 or 8 — Claim 1 - Alkaline earth metal chloride: calcium chloride or magnesium chloride — Claims 1-2 - Catalyst metals: molybdenum, tungsten, osmium or ruthenium — Claim 3

- NBR feed Mooney viscosity: less than 30 Mooney units — Detailed description - NBR feed number-average molecular weight: less than 70000 g/mol — Detailed description - HNBR product molecular weight (Mw): 30000 to 250000 — Detailed description - HNBR product Mooney viscosity (ML 1+4 at 100°C): 3 to 50 — Detailed description - HNBR product polydispersity index: less than 2.5 — Detailed description - HNBR product residual double bond content: low residual double bond contents (RDB) — Detailed description

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate relative to the salt-free control. - Example 8: CsBr also accelerated ring-closing metathesis, though with a different conversion profile over time.

Technical Relevance This patent is relevant because it addresses catalyst-system enhancement for metathesis, a key precursor step for lowering NBR molecular weight before hydrogenation. The disclosed salt-additive strategy is directly applicable to controlling NBR feedstock properties for HNBR manufacture.

Patent 4

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight HNBR produced by metathesis degradation of NBR followed by hydrogenation. The process uses solution-phase metathesis with a co-olefin and ruthenium/osmium metathesis catalysts to reduce molecular weight and narrow the molecular-weight distribution before or during hydrogenation. - Rubber cement concentration: 15 wt.-% solid — Example 1 - Rubber cement concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - 1-hexene amount: 150 g — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity of starting polymer: 73 ML 1+4 @100°C — Table 2 - Mooney viscosity of substrate: 73 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 1 product: 43 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 2 product: 28 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 3 product: 3 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 4 product: 31 ML 1+4 @100°C — Table 2 - Mw of starting polymer: 296000 — Table 2 - Mw of Experiment 1 product: 189000 — Table 2 - Mw of Experiment 2 product: 136000 — Table 2 - Mw of Experiment 3 product: 59000 — Table 2 - Mw of Experiment 4 product: 111000 — Table 2 - PDI of starting polymer: 3.50 — Table 2 - PDI of Experiment 1 product: 2.59 — Table 2 - PDI of Experiment 2

product: 2.27 — Table 2 - PDI of Experiment 3 product: 1.90 — Table 2 - PDI of Experiment 4
product: 2.02 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-based metathesis conditions were used to prepare a lower-viscosity HNBR product. - Example 2: Increasing catalyst loading further reduced Mooney viscosity and molecular weight. - Example 3: Lower solids concentration and metathesis conditions produced the lowest Mooney viscosity and narrowest PDI. - Example 4: 1-hexene was used as co-olefin, demonstrating an alternative chain-scission route with retained low molecular weight.

Technical Relevance This is a core HNBR synthesis patent because it directly teaches controlled molecular-weight reduction of NBR followed by hydrogenation to HNBR. It provides concrete process windows for catalyst loading, temperature, co-olefin choice, and resulting Mooney/Mw targets.

Patent 5

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is a continuation of the low-molecular-weight HNBR metathesis/hydrogenation route. It discloses solution metathesis of NBR with co-olefins under ruthenium-based catalysis, followed by hydrogenation to obtain HNBR with reduced molecular weight and improved processability. - Rubber cement concentration: 15 wt.-% solid — Example 1 - Rubber cement concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - 1-hexene amount: 150 g — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mooney viscosity of starting polymer: 73 ML 1+4 @100°C — Table 2 - Mooney viscosity of substrate: 73 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 1 product: 43 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 2 product: 28 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 3 product: 3 ML 1+4 @100°C — Table 2 - Mooney viscosity of Experiment 4 product: 31 ML 1+4 @100°C — Table 2 - Mw of starting polymer: 296000 — Table 2 - Mw of Experiment 1 product: 189000 — Table 2 - Mw of Experiment 2 product: 136000 — Table 2 - Mw of Experiment 3 product: 59000 — Table 2 - Mw of Experiment 4 product: 111000 — Table 2 - PDI of starting polymer: 3.50 — Table 2 - PDI of Experiment 1 product: 2.59 — Table 2 - PDI of Experiment 2 product: 2.27 — Table 2 - PDI of Experiment 3 product: 1.90 — Table 2 - PDI of Experiment 4 product: 2.02 — Table 2

Relevant Experimental Evidence - Example 1: Ethylene-based metathesis conditions gave a lower-viscosity HNBR product. - Example 2: Higher catalyst loading improved molecular-weight reduction. - Example 3: The lowest Mooney viscosity and PDI were achieved under the most aggressive metathesis conditions. - Example 4: 1-hexene co-olefin was demonstrated as an alternative chain-transfer partner.

Technical Relevance This patent is highly relevant because it quantifies how metathesis conditions translate into HNBR molecular-weight reduction and processability. It complements the earlier patent by showing reproducible low-Mooney HNBR outcomes from controlled degradation/hydrogenation.

Patent 6

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by simultaneously subjecting NBR to metathesis and hydrogenation in the presence of hydrogen and a ruthenium/osmium catalyst. The process is designed to reduce molecular weight and residual double bonds without isolating an intermediate degraded polymer. - HNBR molecular weight (Mw): 20,000 to 250,000 — Detailed description - HNBR Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Detailed description - HNBR polydispersity index: less than 3.0 — Detailed description - NBR feed Mooney viscosity (ML 1+4 @ 100°C): 24 to 70 — Detailed description - Preferred NBR feed Mooney viscosity: 28 to 40 — Detailed description - NBR feed Mw: 200000 to 700000 — Detailed description - Residual double bond content: 1 to 18% — Detailed description - NBR feed Mooney viscosity: less than 30 Mooney units — Detailed description - NBR feed number-average molecular weight: less than 70000 g/mol — Detailed description - Hydrogenation catalyst amount: 0.01 to 1% by weight of polymer — Detailed description - Preferred hydrogenation catalyst amount: 0.03 to 0.5% by weight of polymer — Detailed description - Particularly preferred hydrogenation catalyst amount: 0.1 to 0.3% by weight of polymer — Detailed description - Hydrogenated diene units: 50 to 100% — Detailed description - Preferred hydrogenated diene units: 85 to 100% — Detailed description - Particularly preferred hydrogenated diene units: 95 to 100% — Detailed description

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalysts were compared for hydrogenation efficiency and product properties. - Example 1: HNBR 1 reached RDB 0.4% after 240 min with Mw 113371 and Mooney 33.3. - Example 1: HNBR 2 reached RDB 3.5% after 240 min with Mw 133758 and Mooney 41.3. - Example 1: HNBR 3, using Wilkinson's

catalyst with triphenylphosphine, gave higher residual unsaturation and higher Mooney viscosity than HNBR 1.

Technical Relevance This is one of the most directly relevant patents for HNBR synthesis because it explicitly teaches simultaneous metathesis/hydrogenation to make low-Mooney, low-RDB HNBR in a single step. It provides a clear process strategy for controlling molecular weight and saturation level together.

Patent 7

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a catalyst system for metathesis of nitrile rubber using a ruthenium- or osmium-based carbene catalyst together with an alkaline earth metal chloride. The system is intended to improve metathesis efficiency for NBR degradation, which can then be followed by hydrogenation to HNBR. - Catalyst metal: transition group 6 or 8 — Claim 1 - Alkaline earth metal chloride: calcium chloride or magnesium chloride — Claim 2 - Catalyst metals: molybdenum, tungsten, osmium or ruthenium — Claim 3 - HNBR commercial Mooney viscosity: 55 to 120 — Detailed description - HNBR commercial Mn: about 200000 to 700000 — Detailed description - HNBR PDI: frequently 3 or above — Detailed description - Residual double bond content: 1 to 18% — Detailed description - Lowest workable NBR feed Mooney viscosity: about 30 Mooney units — Detailed description - Hydrogenated nitrile rubber obtained from such feed: about 55 Mooney units — Detailed description - NBR feed Mooney viscosity before hydrogenation: less than 30 Mooney units — Detailed description - NBR feed number-average molecular weight before hydrogenation: less than 70000 g/mol — Detailed description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant because it improves the metathesis catalyst system used to precondition NBR for HNBR synthesis. Better metathesis efficiency can lower catalyst loading and reaction time while preserving the polymer backbone for subsequent hydrogenation.

Patent 8

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by reacting NBR in the presence of hydrogen and a ruthenium- or osmium-based metathesis/hydrogenation catalyst. The process is explicitly aimed at simultaneous hydrogenation and metathesis to obtain HNBR with lower molecular weight and narrower molecular-weight distribution. - HNBR molecular weight (Mw): 20,000 to 250,000 — Detailed description - HNBR Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Detailed description - HNBR polydispersity index: less than 3.0 — Detailed description - Hydrogenated diene units: 50 to 100% — Detailed description - Preferred hydrogenated diene units: 85 to 100% — Detailed description - Particularly preferred hydrogenated diene units: 95 to 100% — Detailed description - NBR feed Mooney viscosity (ML 1+4 @ 100°C): 24 to 70 — Detailed description - Preferred NBR feed Mooney viscosity: 28 to 40 — Detailed description - NBR feed Mw: 200000 to 700000 — Detailed description - Residual double bond content: 1 to 18% — Detailed description - Hydrogenation catalyst amount: 0.01 to 1% by weight of polymer — Detailed description - Preferred hydrogenation catalyst amount: 0.03 to 0.5% by weight of polymer — Detailed description - Particularly preferred hydrogenation catalyst amount: 0.1 to 0.3% by weight of polymer — Detailed description

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson-type catalysts were compared under hydrogenation conditions. - Example 1: HNBR 1 achieved the lowest residual double bond content after 240 min. - Example 1: HNBR 1 showed lower Mooney viscosity and lower Mw than HNBR 3. - Example 1: The catalyst system according to the invention outperformed the industrial norm in hydrogenation efficiency.

Technical Relevance This patent is central to HNBR synthesis because it directly claims simultaneous hydrogenation/metathesis of NBR to low-Mooney HNBR. It provides a practical catalyst framework for producing HNBR with controlled residual unsaturation and molecular weight in one step.

Patent 9

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis; it discloses removal of ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber solutions using functionalized ion-exchange resins. The process is a downstream purification step after metathesis/hydrogenation. - Ruthenium in solution: 5 to 1000 ppm based on optionally hydrogenated nitrile rubber — Claim 2 - Ruthenium in solution: 5 to 500 ppm based on optionally hydrogenated nitrile rubber — Claim 3 - HNBR solution concentration: 0.5 to 30 wt% — Claim 4 - Resin particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Rubber treatment temperature: about 100°C — Example 2A - Treatment time: 66 hours — Example 2A - Initial Ru content: 15 ppm — Table 2 - Ru content after inventive process: 10 ppm — Table 2 - Ru removal: 33% — Table 2

Relevant Experimental Evidence - Example 2A: A functionalized ion-exchange resin reduced Ru content in HNBR solution from 15 ppm to 10 ppm after 66 hours at about 100°C. - Example 2B: The untreated control established the baseline Ru content for comparison.

Technical Relevance This patent is relevant because catalyst residue removal is a critical finishing step for HNBR made by ruthenium-catalyzed metathesis/hydrogenation. It addresses purity, color, and downstream processing issues that arise in HNBR synthesis.

Patent 10

- Patent Number: EP2700692B1
- Patent Title: Hydrogenated nitrile rubber composition
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an HNBR composition for vulcanization-molding, not the synthesis of the polymer itself. It focuses on compounding HNBR with carbon black and/or other fillers, a polyfunctional unsaturated co-crosslinking agent, and peroxide to obtain low-hardness static seals with good LLC resistance and roll processability. - Hydrogenated nitrile rubber: bound acrylonitrile content 31 to 50% — Claim 1 - Hydrogenated nitrile rubber: iodine value 30 mg/100 mg or less — Claim 1 - Carbon black and/or other filler: 30 to 300 parts by weight — Claim 1 - Polyfunctional unsaturated compound cocrosslinking agent: 0.5 to 30 parts by weight — Claim 1 - Organic peroxide: 1 to 10 parts by weight — Claim 1 - Durometer A hardness target: less than 70 — Claim 1 - HNBR-A bound AN content: 36% — Example 1 - HNBR-A iodine value: 18 mg/100 mg — Example 1 - MT carbon black amount: 150 parts by weight — Example 1 - Cocrosslinking agent amount: 6 parts by

weight — Example 1 - Antioxidant amount: 3 parts by weight — Example 1 - Organic peroxide amount: 3 parts by weight — Example 1 - Compression molding temperature: 180°C — Example 1 - Compression molding time: 10 minutes — Example 1 - LLC test temperature: 120°C — Example 1 - LLC test duration: 2000 hours — Example 1 - Mechanical seal service temperature: up to 160°C — Background

Relevant Experimental Evidence - Example 1: A peroxide-cured HNBR sheet was prepared from HNBR-A, MT carbon black, cocrosslinker, antioxidant, and peroxide. - Example 1: The composition was kneaded in a 3 L kneader and on a 10-inch roll, then compression molded at 180°C for 10 minutes. - Example 1: The resulting material was evaluated for LLC resistance, swelling, softening, and roll processability. - Example 15: Combined MT carbon black and silica were used to tune filler balance and performance. - Example 18-19: Silane coupling agent plus mixed fillers were used to further optimize the composition.

Technical Relevance This patent is relevant because it shows how HNBR must be formulated after synthesis to meet demanding seal requirements, especially LLC resistance and low hardness. It provides practical compounding windows that complement HNBR polymer synthesis by defining end-use property targets.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Several patents converge on low-Mooney HNBR targets: US7772328B2 and US7951875B2 report products down to Mooney 3–43, while EP2473281B1 and US9150669B2 claim broader HNBR windows of 1–50 Mooney.
- The synthesis patents split into two routes: sequential metathesis then hydrogenation (US7772328B2, US7951875B2, US7737233B2) versus simultaneous metathesis/hydrogenation in hydrogen (EP2473281B1, US9150669B2).
- Catalyst systems differ: US7737233B2 adds alkaline earth metal chlorides to ruthenium/osmium metathesis catalysts; EP2473281B1 and US9150669B2 use ruthenium/osmium complexes for one-step hydrogenation/metathesis; US8957161B2 emphasizes salt additives to accelerate metathesis.
- Process conditions in the low-MW patents are solution-based, often using co-olefins such as ethylene or 1-hexene, with reactor temperatures around 80°C in the example sets and catalyst loadings from 0.05 to 0.25 phr in US7772328B2/US7951875B2.
- Residual unsaturation control is explicit in EP2473281B1 and US9150669B2, where example products reach RDB values as low as 0.4% after 240 min, whereas the metathesis-only patents focus more on molecular-weight reduction before hydrogenation.
- Downstream formulation patents (US9321910B2, US9243683B2, EP2700692B1) show that HNBR synthesis is only part of the technology stack; peroxide cure, microgels, fillers, and plasticizers are used to tailor heat resistance, cold resistance, stiffness, and LLC resistance.

- US7846995B2 addresses a different but essential manufacturing issue: removal of ruthenium residues from HNBR solutions, with 5–1000 ppm Ru in solution and 33% removal demonstrated in Example 2A.

4. CONCLUSION

The patent set shows that hydrogenated NBR technology is dominated by two synthesis philosophies: controlled NBR metathesis followed by hydrogenation, and true one-step simultaneous metathesis/hydrogenation under hydrogen. The strongest evidence for practical HNBR synthesis supports solution-phase processing, ruthenium-based single-site catalysts, optional co-olefins, and feedstocks with moderate Mooney viscosity so that the final HNBR reaches low Mooney values, low residual double bond content, and narrow molecular-weight distribution. Based on the evidence, the most robust synthesis targets are NBR feed Mooney viscosity around 24 to 40, catalyst loading in the low phr or 0.1 to 0.3 wt% range, reaction temperatures near 80°C in the example sets, and hydrogenation sufficient to drive diene saturation toward 95 to 100% when high-performance HNBR is desired. For downstream use, peroxide cure systems, microgels, and carefully balanced fillers are important to translate the polymer into seals, belts, hoses, and gaskets with the required heat, oil, and coolant resistance. Residue-removal steps should also be considered when ruthenium catalysts are used, because catalyst purification is a practical requirement for commercial HNBR manufacture.

5. REFERENCES

- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
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- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>

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